Unity Catalog

A unified governance solution for all your data, analytics, and AI assets on the Databricks Lakehouse Platform — bringing centralised control, security, and observability to every layer of your data ecosystem.

DATA GOVERNANCE | DATABRICKS | UNITY CATALOG

Why Governance Matters

As data volumes grow and regulatory scrutiny intensifies, uncontrolled data access is no longer acceptable. Organisations must enforce clear boundaries around who can access what — and prove it to auditors.

Data Access Risks

Without governance, sensitive data is exposed to unauthorised users, increasing the risk of breaches, data leaks, and misuse across teams and environments.

Compliance Requirements

Regulations such as GDPR, HIPAA, and SOC 2 mandate strict data handling controls. Governance frameworks provide the audit trails and access logs needed to demonstrate compliance.

What is Unity Catalog?

A Centralised Metadata Layer

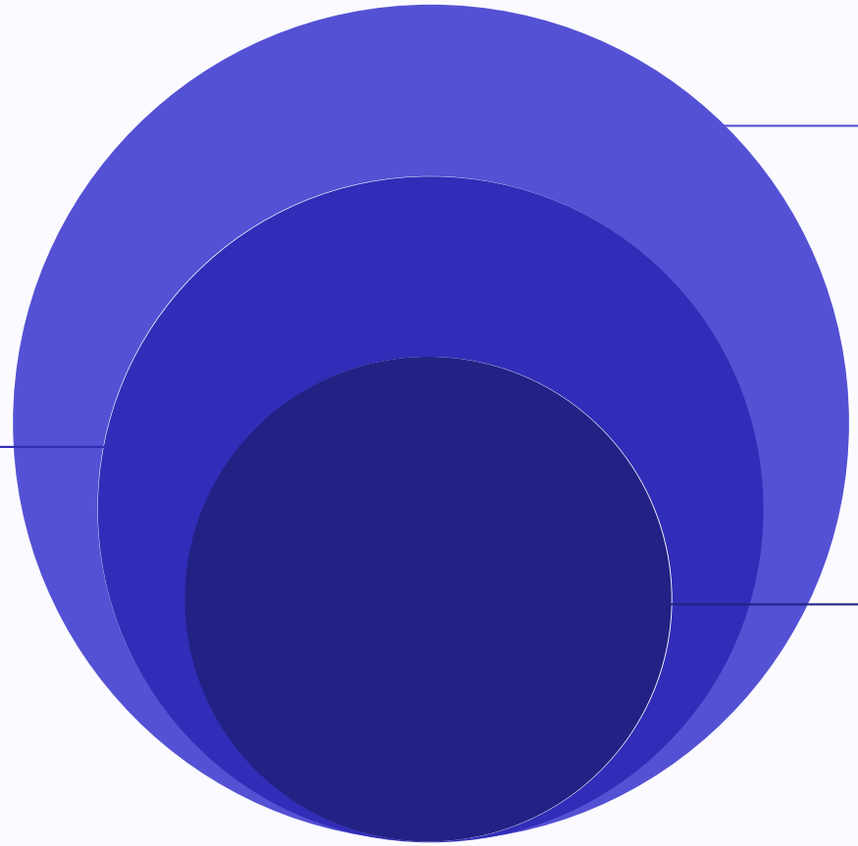
Unity Catalog provides a single pane of glass for discovering, managing, and auditing all data assets — tables, views, files, models, and dashboards — across your entire Databricks workspace and beyond.

A Unified Governance System

It enforces consistent security policies, access controls, and lineage tracking at scale, replacing fragmented, workspace-by-workspace configurations with one coherent governance framework.

The Unity Catalog Hierarchy

Unity Catalog organises all data assets within a clear, three-level namespace. This structure enables precise, scoped governance policies at every level of granularity.



This hierarchy maps naturally to real-world organisational structures — for example, a **Finance** catalogue containing a **Reporting** schema, which holds individual **Revenue** and **Expense** tables. Policies applied at the catalogue level cascade downward, reducing administrative overhead.

Access Control with RBAC

Unity Catalog uses Role-Based Access Control (RBAC) to define and enforce who can interact with which data assets. Permissions are granted or revoked using standard SQL-compatible commands, making them easy to audit and automate.

GRANT

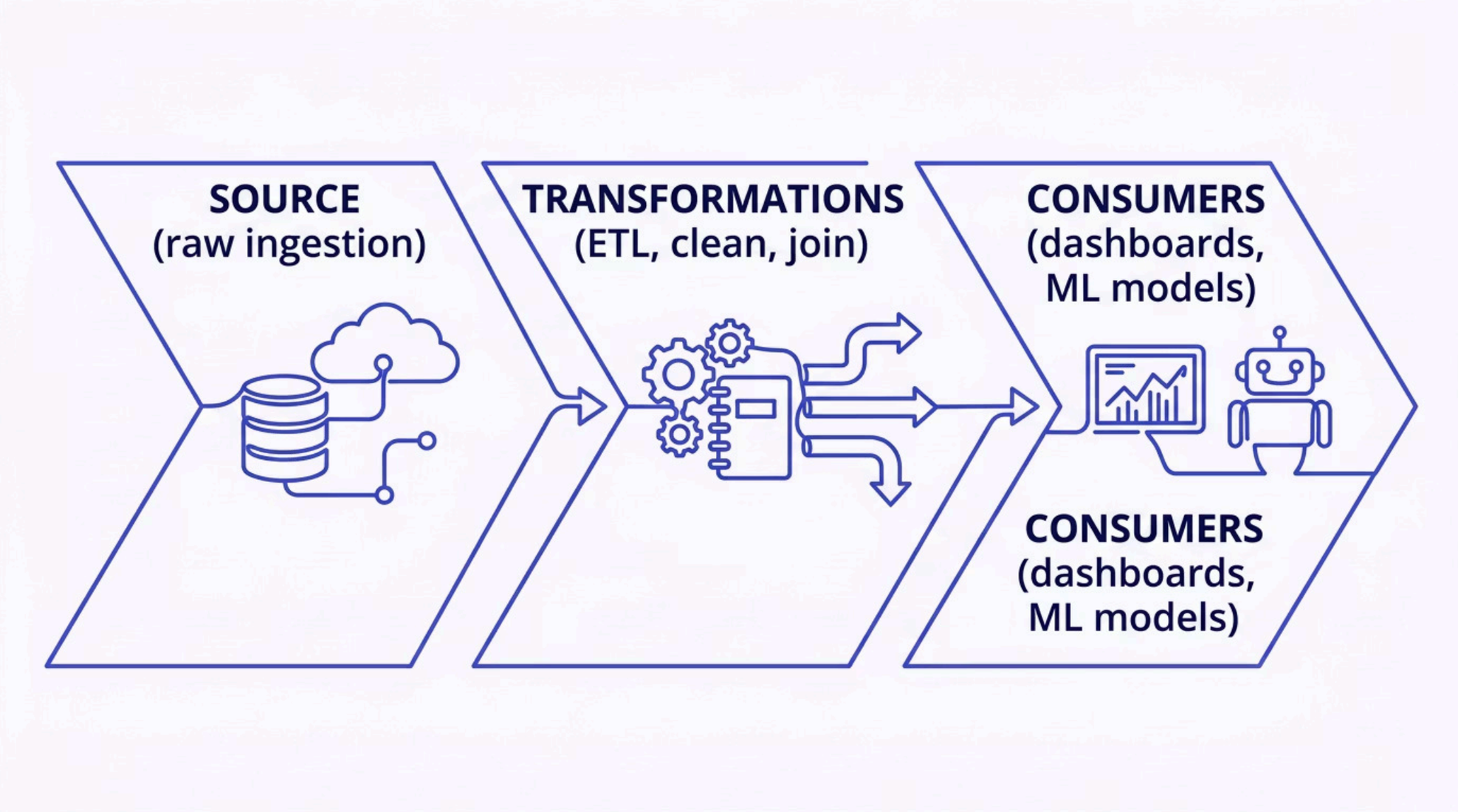
Use `GRANT` to assign specific privileges — such as `SELECT`, `MODIFY`, or `CREATE TABLE` — to a user, group, or service principal on a catalogue, schema, or table. Permissions can be scoped precisely, ensuring users access only what they need.

REVOKE

Use `REVOKE` to remove previously granted privileges. This is essential for offboarding team members, rotating service accounts, or tightening access controls as data sensitivity increases over time.

Data Lineage

Unity Catalog automatically captures end-to-end data lineage — no manual tagging required. Understanding exactly how data flows through your platform is critical for debugging pipelines, assessing impact, and satisfying regulatory requirements.



With lineage visibility, data stewards can instantly answer: *"Where did this number come from?"* and *"Who will be affected if I change this table?"* — reducing risk and accelerating root cause analysis.

Integration with Delta Lake

Table-Level Governance

Unity Catalog sits natively on top of Delta Lake, the open-source storage layer at the heart of the Databricks Lakehouse. Every Delta table registered in Unity Catalog inherits full governance capabilities — access control, lineage, auditing, and versioning — without any additional configuration.

- Unified permissions across all workspaces
- Full audit log of all read and write operations
- Consistent policy enforcement across batch and streaming workloads

Column-Level Security

Beyond table-level access, Unity Catalog enables fine-grained column masking and row-level filters. Sensitive fields such as PII, financial data, or health records can be masked or restricted for specific roles — without duplicating or transforming the underlying dataset.

- Dynamic data masking for PII fields
- Row filters based on user group or region
- No data duplication required for access tiers

Real-World Access Control Example

Consider a typical enterprise scenario where the same data platform serves multiple teams with very different access requirements. Unity Catalog enforces the right boundaries automatically.

Finance Team — Restricted Access

Finance analysts are granted full `SELECT` access to the `finance` catalogue, including sensitive revenue, payroll, and cost tables. Access is restricted to named users and finance service principals only. All queries are logged for SOX compliance auditing.

Marketing Team — Limited Access

Marketing users can access aggregated campaign performance tables within the `marketing` schema, but are blocked from any financial or HR data. Column masking is applied to any customer PII fields that appear in shared datasets, protecting privacy by default.

Best Practices for Unity Catalog

Implementing Unity Catalog successfully requires more than just configuration — it demands deliberate design decisions from day one. These principles will help you build a governance model that scales with your organisation.

1 Separate Environments with Distinct Catalogues

Maintain separate `dev`, `staging`, and `prod` catalogues to prevent accidental modifications to production data during development or testing. Permissions, data quality, and lineage are tracked independently per environment, reducing cross-environment risk significantly.

2 Enforce the Principle of Least Privilege

Grant users and service principals only the minimum permissions required to perform their role. Start with read-only access and escalate deliberately. Regularly audit active grants and revoke any that are no longer needed. This limits blast radius in the event of a compromised account or misconfiguration.

You Now Have the Full Picture

Pipeline + Governance, end to end.

From raw ingestion to governed, production-ready data — you now have the tools and knowledge to build a trustworthy, scalable data platform on Databricks.

What You've Covered

01

Why governance matters

Access risks, compliance, and the cost of doing nothing

02

Unity Catalog architecture

Centralised metadata, three-level hierarchy, RBAC

03

Lineage, Delta integration & real examples

End-to-end observability with fine-grained security

04

Best practices to operationalise

Environments, least privilege, and scalable policy design